Predictive Analytics and Risk Stratification

ANL 220

Predictive analytics uses data to estimate the likelihood of future events or outcomes. In public health, predictive models may be used to anticipate disease burden, identify populations at higher risk, prioritize outreach, allocate limited resources, or support early intervention. Risk stratification is the practice of grouping individuals, facilities, areas, events, or populations into categories based on estimated risk. This module helps learners distinguish prediction from diagnosis, causation, eligibility determination, and public health judgment. A risk score can support planning or triage, but it should not automatically become a decision without validation, review, and governance. Public health staff need to know how predictive models are created, what performance measures mean, and how errors affect equity, trust, and operations. By the end of this module, learners should be able to review a predictive AI proposal, identify the data and outcome being predicted, assess the consequences of false positives and false negatives, and design appropriate human review and monitoring processes.

Level	applied/foundational
Primary track	Analytics and Modeling Track
Appears in tracks	analytics-modeling

Learning Objectives

- Define predictive analytics and risk stratification in public health terms.
- Distinguish prediction from causation, diagnosis, and eligibility determination.
- Interpret basic risk categories, thresholds, calibration, and error tradeoffs.
- Identify equity and operational risks associated with risk scores.
- Design safeguards for use of predictive outputs in resource allocation, outreach, surveillance, or triage.
- Produce a predictive model review plan for a public health workflow.

Module Overview

Predictive analytics uses data to estimate the likelihood of future events or outcomes. In public health, predictive models may be used to anticipate disease burden, identify populations at higher risk, prioritize outreach, allocate limited resources, or support early intervention. Risk stratification is the practice of grouping individuals, facilities, areas, events, or populations into categories based on estimated risk.

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Why This Topic Matters for Public Health AI

Predictive analytics matters because public health agencies often must act before all information is available. During outbreaks, emergencies, or resource-constrained programs, agencies may need to decide where to send staff, which facilities need follow-up, which communities may need outreach, or which reports require expedited review. Prediction can help organize uncertainty, but it cannot remove uncertainty.

Risk stratification can also create harm if used carelessly. A model may assign lower risk to groups with incomplete data, limited healthcare access, or fewer documented encounters. It may assign higher risk based on factors that reflect social vulnerability without ensuring supportive intervention. If a score is used to deny services, delay review, or reduce attention, the tool may worsen inequities.

The responsible use of prediction requires a clear purpose, well-defined outcome, appropriate data, subgroup evaluation, calibrated scores, human review, and transparency about limitations. Public health staff should ask who benefits from the prediction, who could be harmed, and whether the intended action is supportive, punitive, or resource-constraining.

Public Health Example

A local health department may want to predict which neighborhoods are at increased risk of low vaccination coverage before school entry. The model could use prior immunization data, school enrollment patterns, appointment availability, language access indicators, transportation barriers, and historical outreach response. The output might help prioritize community outreach and mobile clinics.

This use can support equity if it directs resources toward communities facing access barriers. It can create harm if the score stigmatizes neighborhoods, ignores missing data, or shifts responsibility away from system-level barriers.

The model should be evaluated for calibration, data completeness, community context, and whether the intervention is supportive and accessible.

Technical Deep Dive

A predictive workflow begins with the outcome. The outcome must be specific, measurable, and aligned with a public health action. Predicting “risk” is not enough. Staff must define risk of what, during what time period, for which population, and for what purpose. A model predicting hospitalization within 30 days is different from one predicting delayed case investigation or low service uptake.

Feature selection should be reviewed for both technical and ethical implications. Some features may improve prediction but create fairness, privacy, or interpretability concerns. Geography, insurance status, prior utilization, language, disability, housing instability, or justice involvement may reflect structural inequities. These features require careful review because they may be useful for supportive outreach but inappropriate for restrictive decisions.

Calibration and discrimination answer different questions. Discrimination describes whether a model can separate higher-risk from lower-risk cases. Calibration describes whether predicted probabilities correspond to observed outcomes. A model can rank cases reasonably well while still overstating or understating absolute risk, which matters when decisions depend on risk categories or thresholds.

Thresholds are policy and workflow decisions as much as technical decisions. A lower threshold may catch more true high-risk cases but create more false positives and burden staff. A higher threshold may conserve resources but miss people or places needing support. Threshold selection should consider staffing capacity, consequence severity, equity, and the availability of human review.

Post-deployment monitoring is essential. A model that worked during development may degrade as disease patterns, reporting systems, access to care, community behavior, or program operations change. Monitoring should include performance, calibration, subgroup effects, workflow impact, user overrides, and community feedback.

Risks, Failure Modes, and Guardrails

Prediction used as decision. A risk score may be treated as a final decision rather than decision support. Guardrails include role-based guidance, human review, and documentation of final decisions.

Equity distortion. Risk models may reflect unequal data availability or historical inequities. Guardrails include subgroup analysis, community review, and restrictions on punitive uses.

Threshold harm. Poorly selected thresholds can create alert fatigue or missed risk. Guardrails include threshold review, scenario testing, and monitoring of false positives and false negatives.

Model drift. Performance may change over time. Guardrails include periodic recalibration, drift monitoring, and authority to pause or modify use.

Application to AI-Supported Workflows

Predictive analytics may be embedded in dashboards, case management systems, early warning tools, outreach lists, or agentic workflows. Generative AI may explain or summarize predictive outputs, but those explanations must not imply certainty that the model does not provide. RAG systems may combine risk results with policy guidance, while agentic workflows may trigger tasks or notifications based on thresholds.

When prediction supports public health action, the workflow should make the score, its limitations, and the source data visible to users. Staff should know when a score is advisory, when it requires review, and when it cannot be used as the sole basis for action.

Reflection Questions

What outcome is being predicted, and over what time period?

What public health action will follow a high-risk score?

Which data sources may underrepresent some populations?

What are the consequences of false positives and false negatives?

How will thresholds, calibration, and subgroup performance be monitored?

Practical Exercise

Create a predictive model review plan for one public health use case. Include the outcome, population, time horizon, data sources, candidate features, risk categories, thresholds, performance measures, subgroup evaluation plan, human review process, and monitoring indicators.

The plan should also identify whether the use is supportive, restrictive, or consequential. Predictive tools used to allocate supportive resources have different risk profiles than tools used to deny, delay, or reduce services.

Expected Artifact or Evidence

Predictive model review plan

Risk score interpretation guide

Threshold and error tradeoff memo

Subgroup monitoring plan

Expected Assignment

- Predictive model review plan
- Risk score interpretation guide
- Threshold and error tradeoff memo
- Subgroup monitoring plan

Knowledge Check

- 1. What is calibration?
- 2. Why should thresholds be reviewed by public health staff?
- 3. Which use of a risk score is most concerning without additional safeguards?
- 4. What is a false negative in a predictive workflow?
- 5. What should a predictive model review plan include?

References and Resources

- NIST Artificial Intelligence Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- NIST AI RMF Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- CDC Evaluation Framework: https://www.cdc.gov/evaluation/
- CDC Public Health Data Strategy: https://www.cdc.gov/public-health-data-strategy/php/about/index.html